Final Capstone Project Codes – BookMyShow CI/CD & Deployment

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Batch 1

Jenkinsfile:

```
pipeline {
  agent any

  tools {
    jdk 'JAVA_HOME'
    nodejs 'NodeJS'
  }

  environment {
    SCANNER_HOME = tool 'sonar-scanner'              // Jenkins > Tool
  }

  stages {
    stage('Clean Workspace') {
      steps {
        cleanWs()
      }
    }
    stage('Checkout from Git') {
      steps {
        git branch: 'feature', url: 'https://github.com/bimalk07/Book-My-Show.git'
        sh 'ls -la'                                  // Verify files after checkout
      }
    }
    stage('SonarQube Analysis') {
      steps {
        withSonarQubeEnv('sonar') {                  // Jenkins > System
          sh '''
            $SCANNER_HOME/bin/sonar-scanner \
            -Dsonar.projectName=bimal-sonar-project \
            -Dsonar.projectKey=bimal-sonar-project
          '''
        }
      }
    }
    stage('Quality Gate') {
      steps {
        timeout(time: 2, unit: 'MINUTES') {
```

```
            waitForQualityGate abortPipeline: true
        }
      }
    }
    stage('Install Dependencies') {
      steps {
        sh '''
          cd bookmyshow-app
          ls -la                                                # Verify package.json exists
          if [ -f package.json ]; then
            rm -rf node_modules package-lock.json       # Remove old dependencies
            npm i chokidar
            npm install                                         # Install fresh dependencies
          else
            echo "Error: package.json not found in bookmyshow-app!"
            exit 1
          fi
        '''
      }
    }

    stage('Docker Build & Push') {
      steps {
        script {
          withDockerRegistry(credentialsId: 'docker-cred') {
            sh '''
              echo "Building Docker image..."
              docker build --no-cache -t bimalk074/bms:latest -f bookmyshow-
app/Dockerfile bookmyshow-app

              echo "Pushing Docker image to registry..."
              docker push bimalk074/bms:latest
            '''
          }
        }
      }
    }
    stage('Deploy to Container') {
      steps {
        sh '''
          echo "Stopping and removing old container..."
          docker stop bms || true
          docker rm bms || true

          echo "Running new container on port 3000..."
          docker run -d --restart=always --name bms -p 3001:3000 bimalk074/bms:latest
```

```
                echo "Checking running containers..."
                docker ps -a

                echo "Fetching logs..."
                sleep 5                         # Give time for the app to start
                docker logs bms
            '''
        }
    }
}
post {
    success {
        emailext (
            subject: "SUCCESS: SonarQube Quality Gate Passed",
            body: """
                Hi Team,

                The SonarQube quality gate passed successfully.

                You can view the report here:
                http://13.58.183.234:9000/dashboard?id=bimal-sonar-project
                username: admin, passwd: admin@Sonar10

                If you want to download a PDF report, please use the SonarQube interface or a
PDF plugin.

                Regards,
                Jenkins
            """,
            to: 'soapmac055@gmail.com'
        )
    }

    failure {
        emailext (
            subject: "FAILED: SonarQube Quality Gate",
            body: """
                Hi Team,

                The SonarQube quality gate has failed.

                Please check the details:
                http://13.58.183.234:9000/dashboard?id=bimal-sonar-project
                username: admin, passwd: admin@Sonar10

                Regards,
                Jenkins
```

```
            """,
            to: 'soapmac055@gmail.com'
        )
      }
   }
}
```

Dockerfile

```
# Use Node.js 18 (or your Jenkins-configured version)
FROM node:18

# Set working directory
WORKDIR /app

# Copy package.json and package-lock.json
COPY package.json package-lock.json ./

# Force install a compatible PostCSS version to fix the issue
RUN npm install postcss@8.4.21 postcss-safe-parser@6.0.0 --legacy-peer-deps

# Install dependencies
RUN npm install chokidar
RUN npm install

# Copy the entire project
COPY . .

# Expose port 3000
EXPOSE 3000

# Set environment variable to prevent OpenSSL errors
ENV NODE_OPTIONS=--openssl-legacy-provider
ENV PORT=3000

# Start the application
CMD ["npm", "start"]
```

Deployment.yml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: bms-app
  namespace: bimal
spec:
  replicas: 2
  selector:
```

```
    matchLabels:
      app: bms
  template:
    metadata:
      labels:
        app: bms
    spec:
      containers:
      - name: bms-container
        image: bimalk074/bms:latest      # Replace with your Docker image
        ports:
        - containerPort: 3000            # Replace with the port your app runs on
```

Service.yml

```
apiVersion: v1
kind: Service
metadata:
  name: bms-service
  namespace: bimal
spec:
  type: LoadBalancer
  ports:
  - port: 80
    targetPort: 3000          # Replace with the port your app runs on
  selector:
    app: bms
```